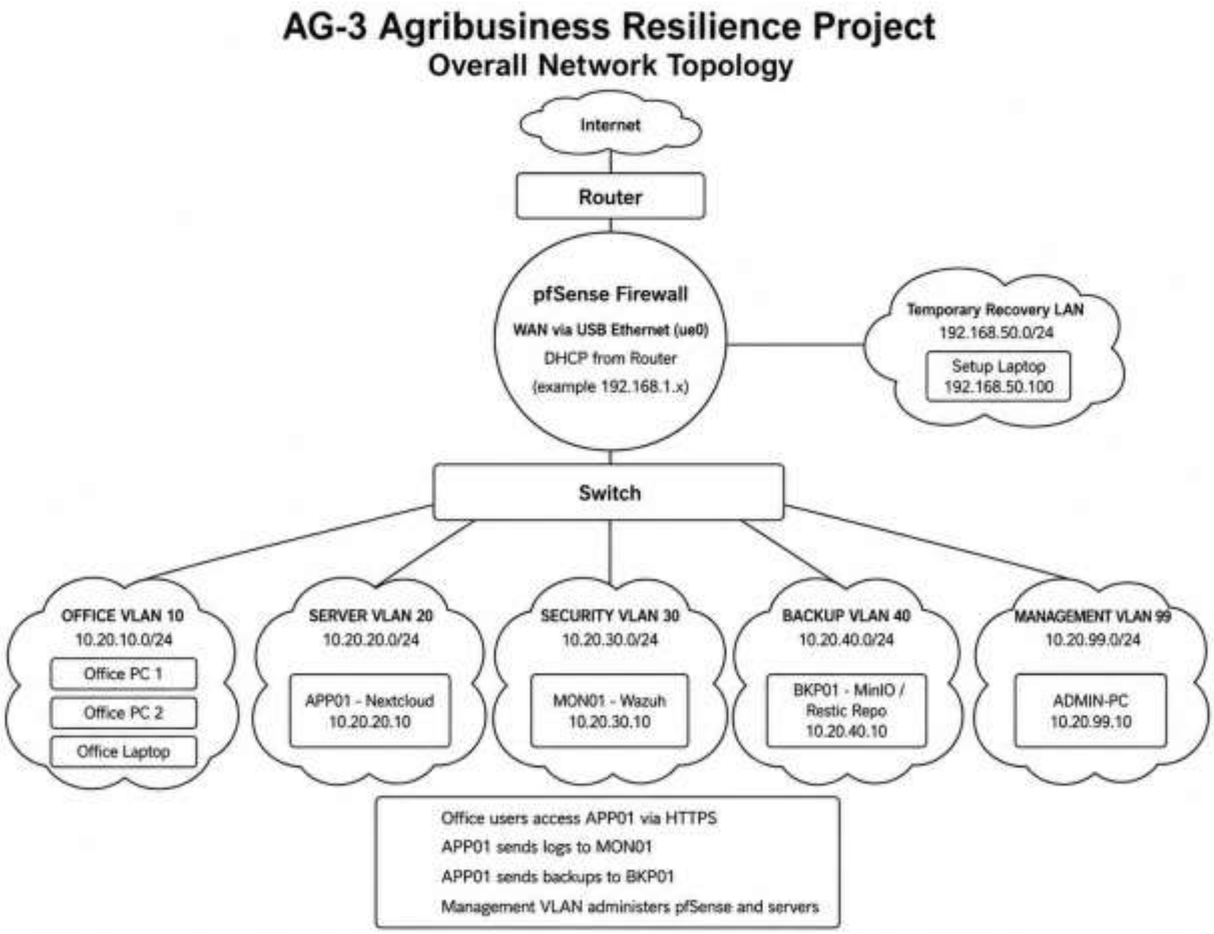
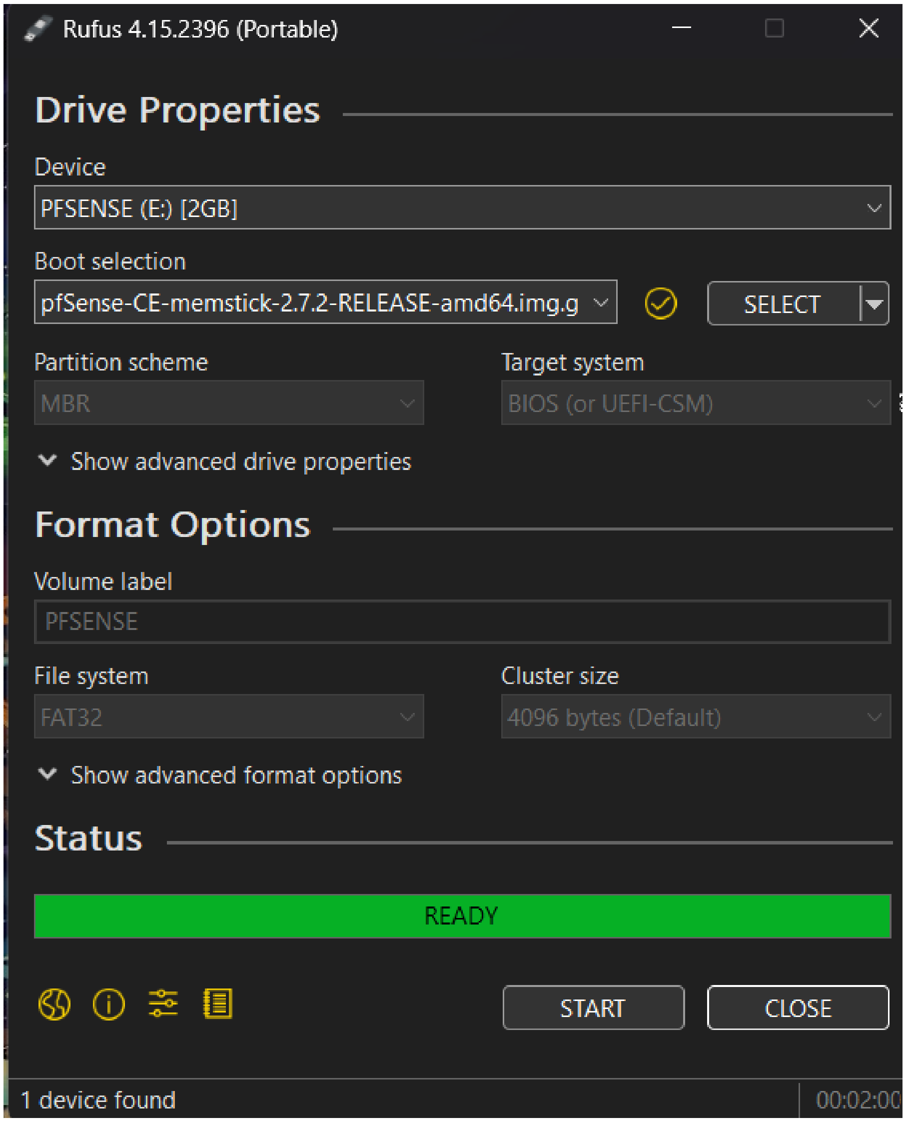
The network topology for AG-3 Agribusiness Resilience Project showing the pfSense firewall, VLAN-based network segmentation, protected business server, Wazuh security monitoring, isolated backup system, and management network.

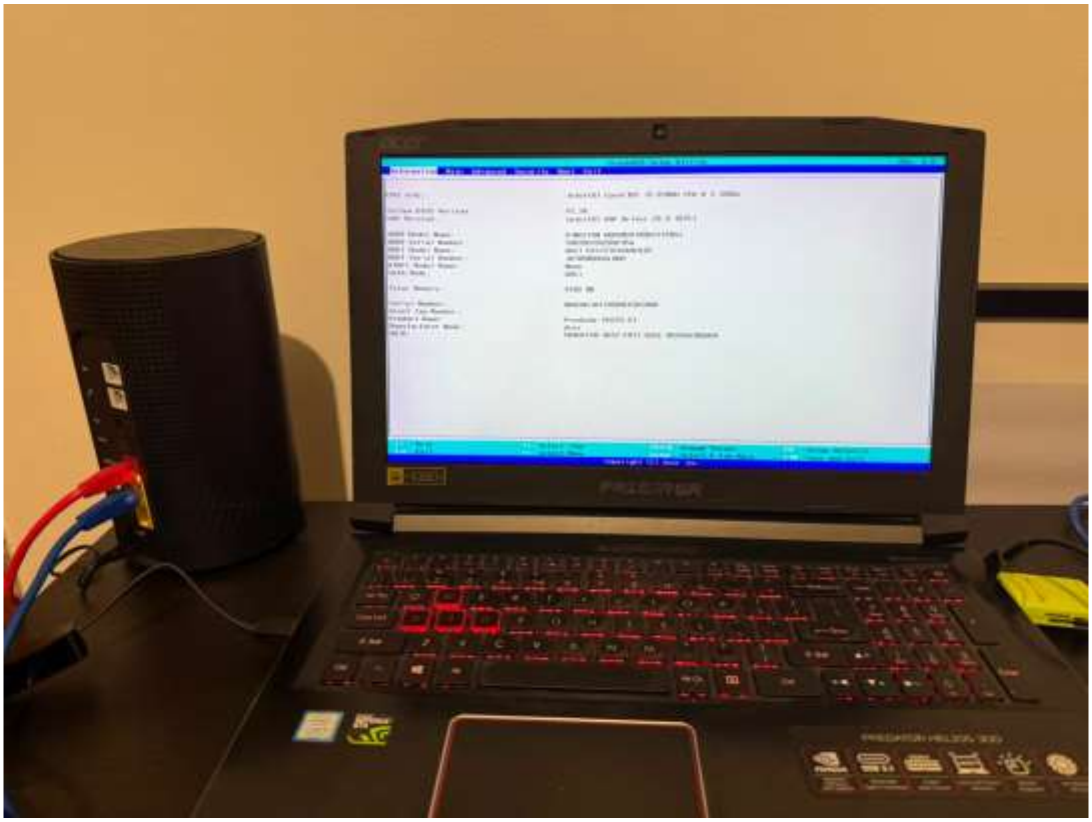


How I set up and configure the pfsense firewall:

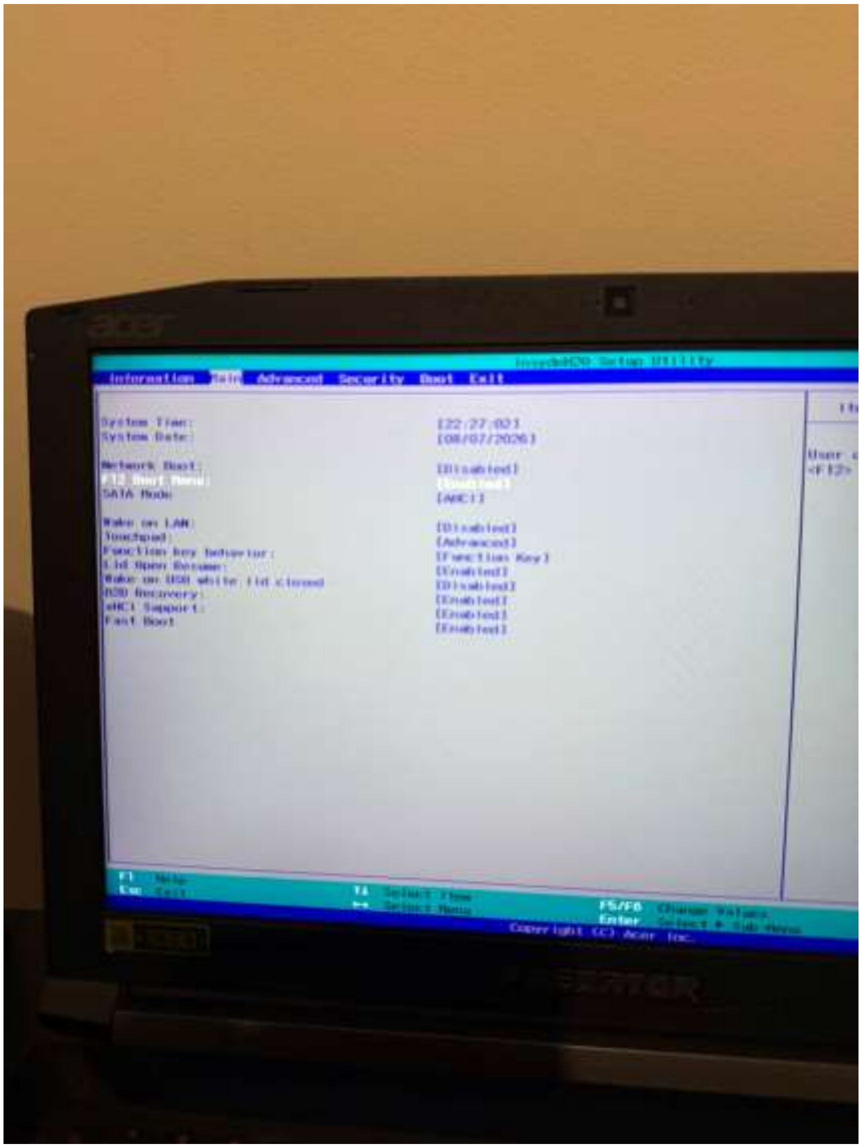
I used Rufus to burn pfSense-CE-memstick-2.7.2-RELEASE-amd64.img into a memory stick to make it bootable for the PC that I used as the firewall.



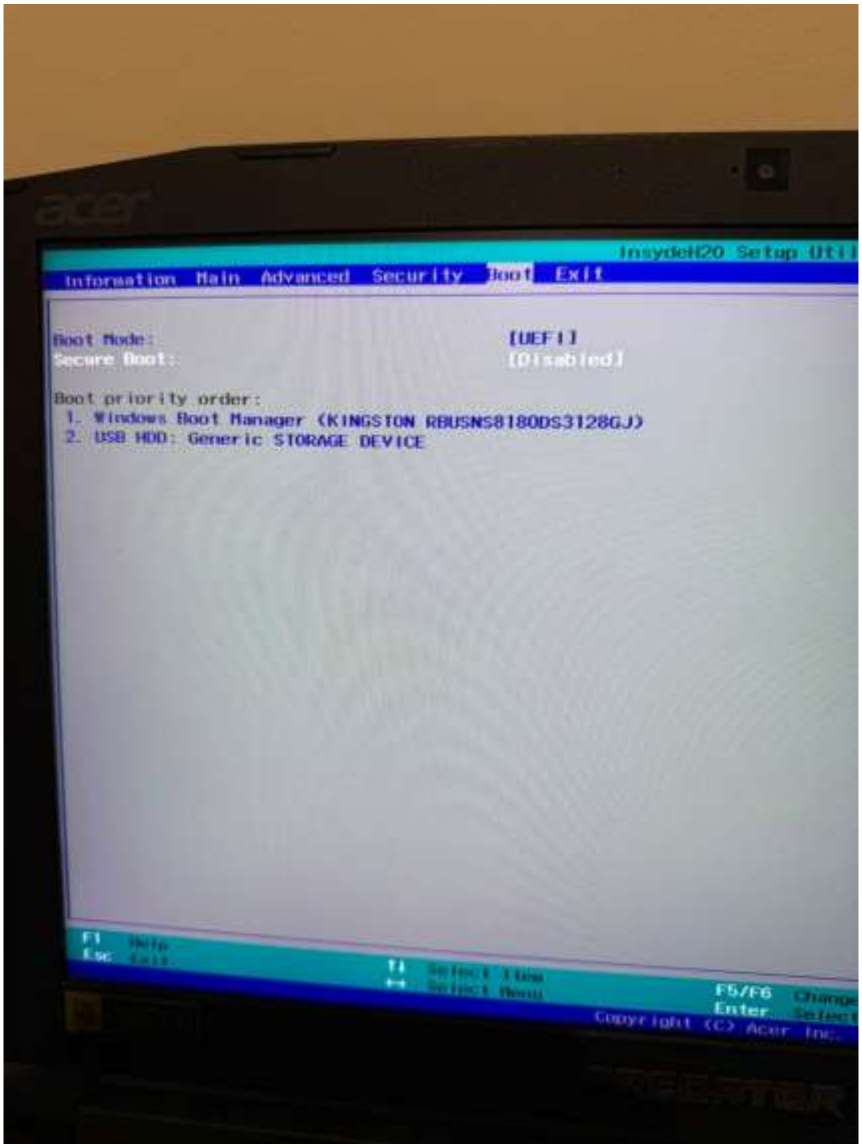
After making the pfSense USB bootable, I checked the firewall PC hardware and BIOS first. The PC has an Intel Core i5-8300H processor, 8 GB memory and a Kingston SSD that I used for the pfSense installation.



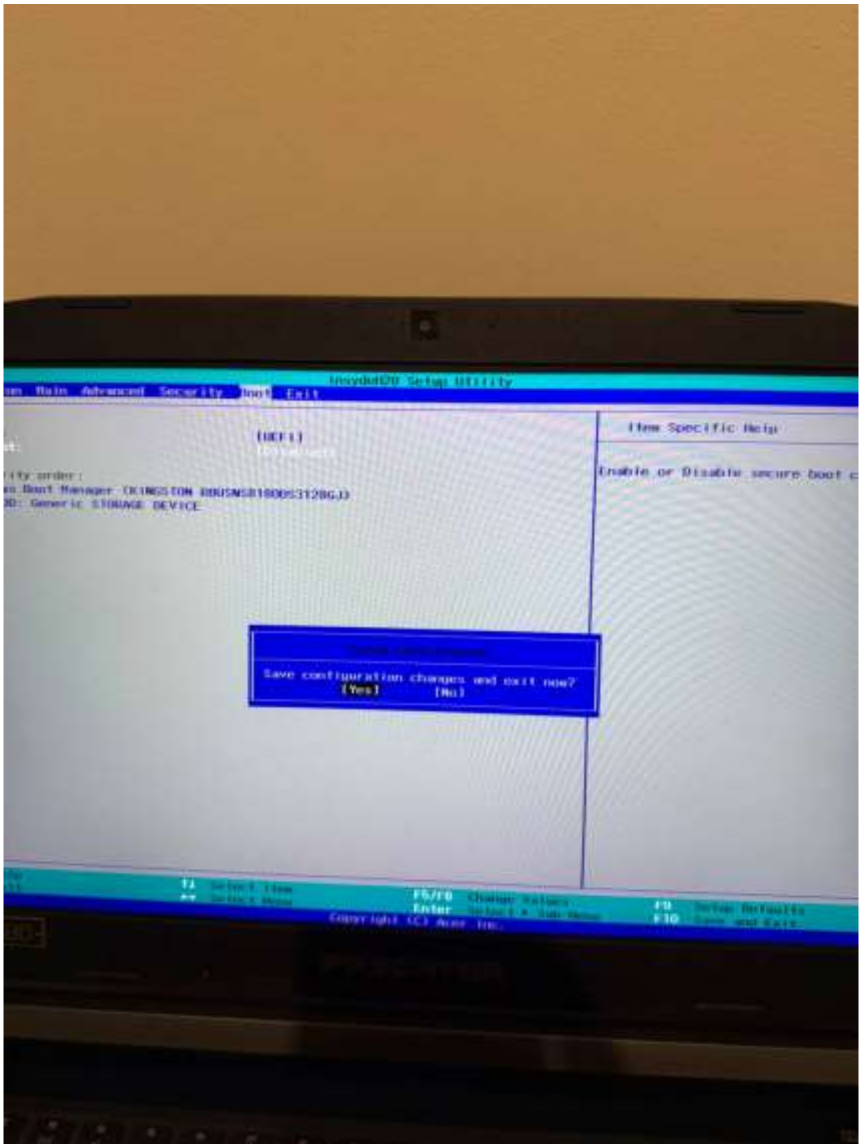
Next, I checked the BIOS settings and enabled the F12 Boot Menu. This made it easier for me to select the bootable USB without changing the normal boot process permanently.



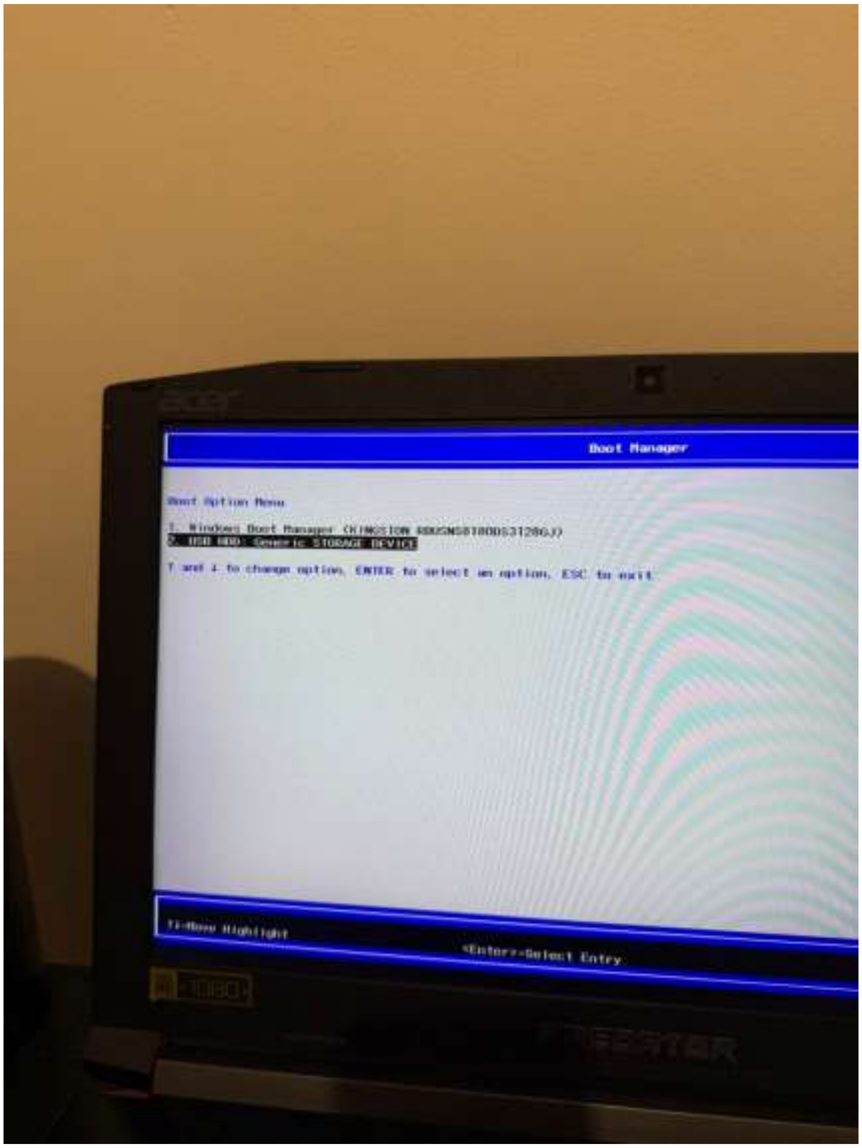
In the Boot settings, I kept the system in UEFI mode and Secure Boot was disabled. I then prepared the laptop to boot the pfSense installer from the USB device.



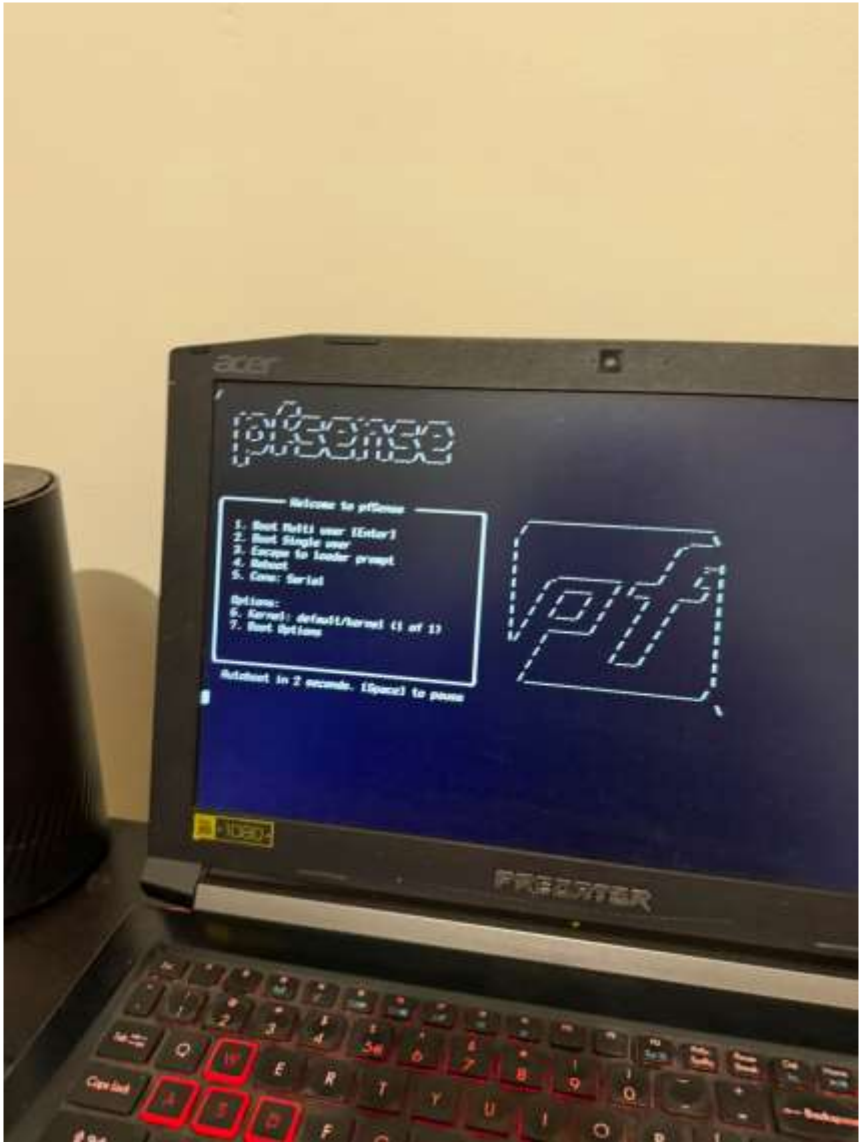
After changing the required BIOS options, I saved the configuration and exited the BIOS.



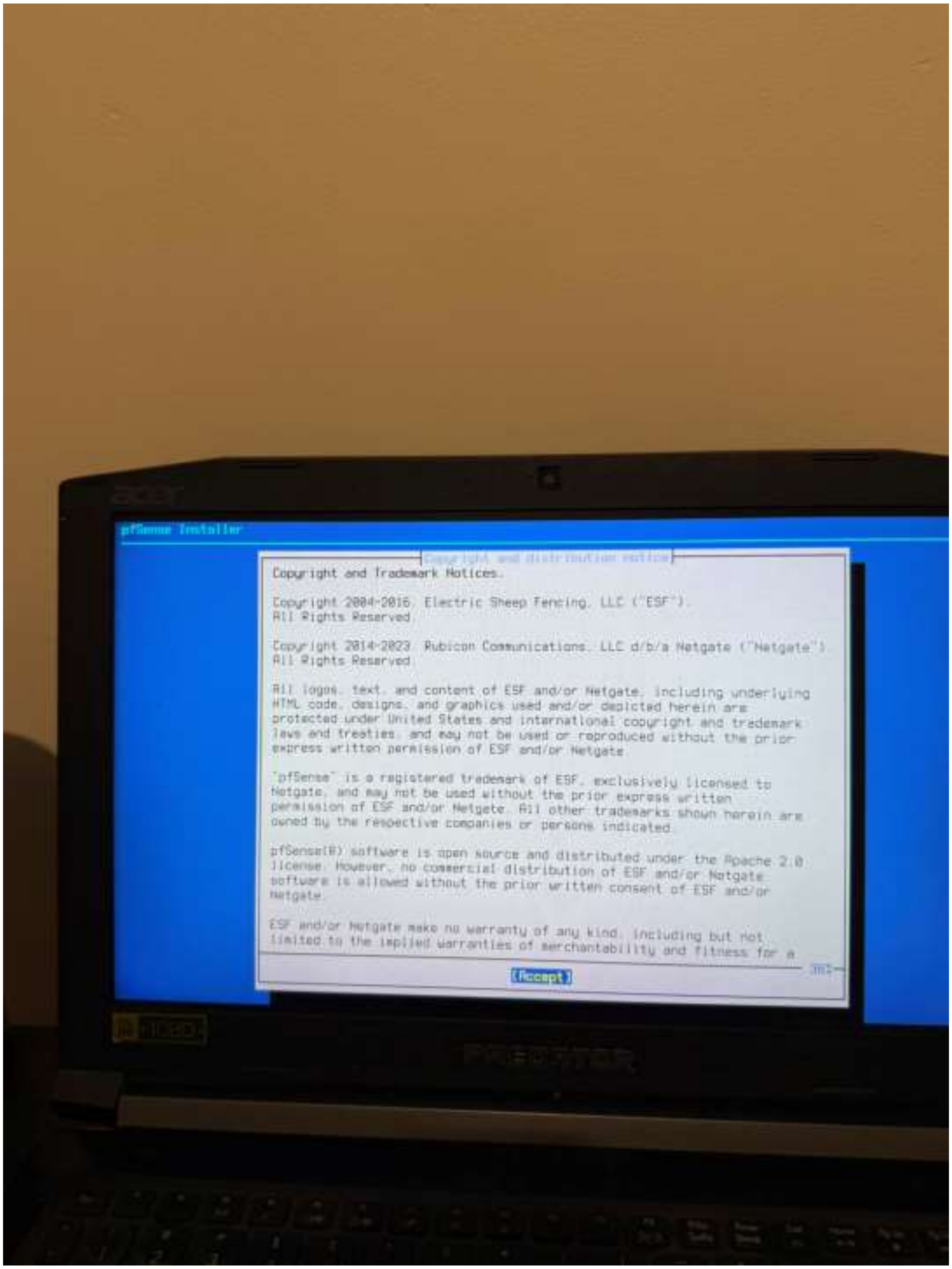
Then, from the Boot Manager, I selected the USB HDD Generic STORAGE DEVICE. This started the pfSense installation media that I created earlier with Rufus.



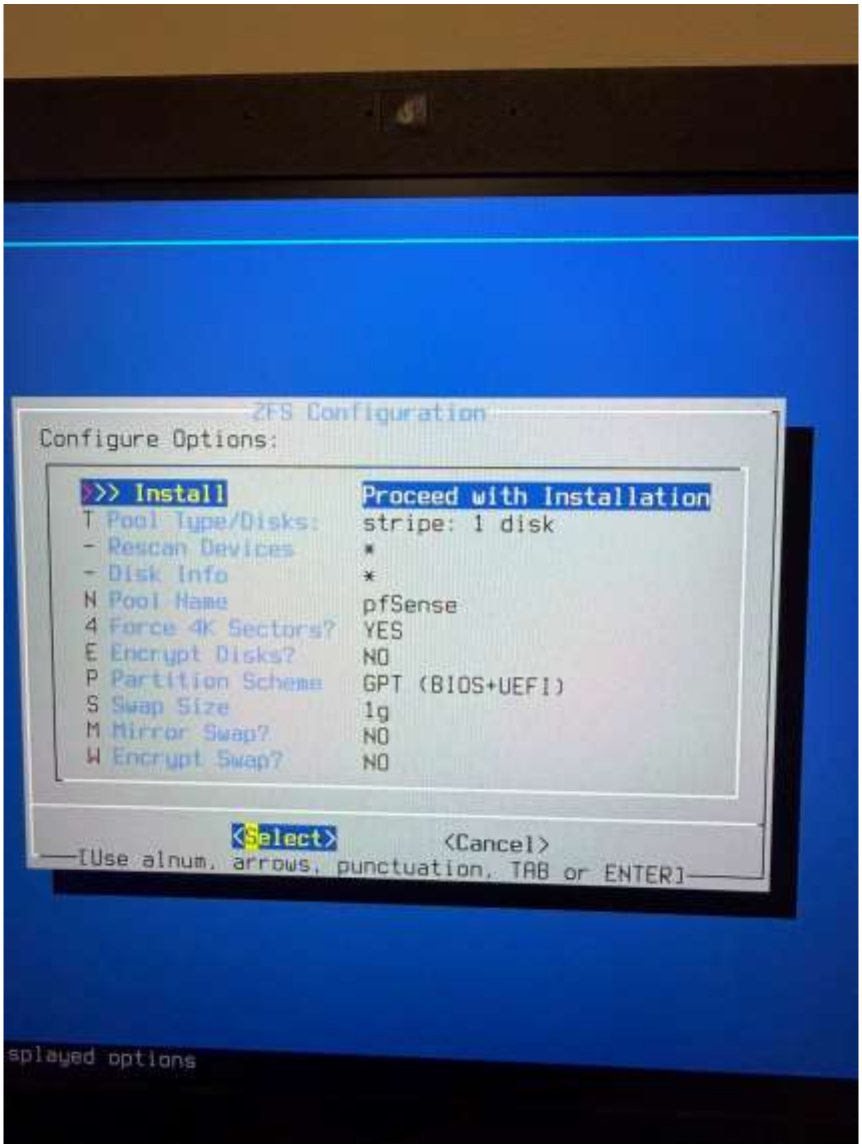
The pfSense boot menu loaded successfully from the USB. I used the normal multi-user boot option to continue the installation.



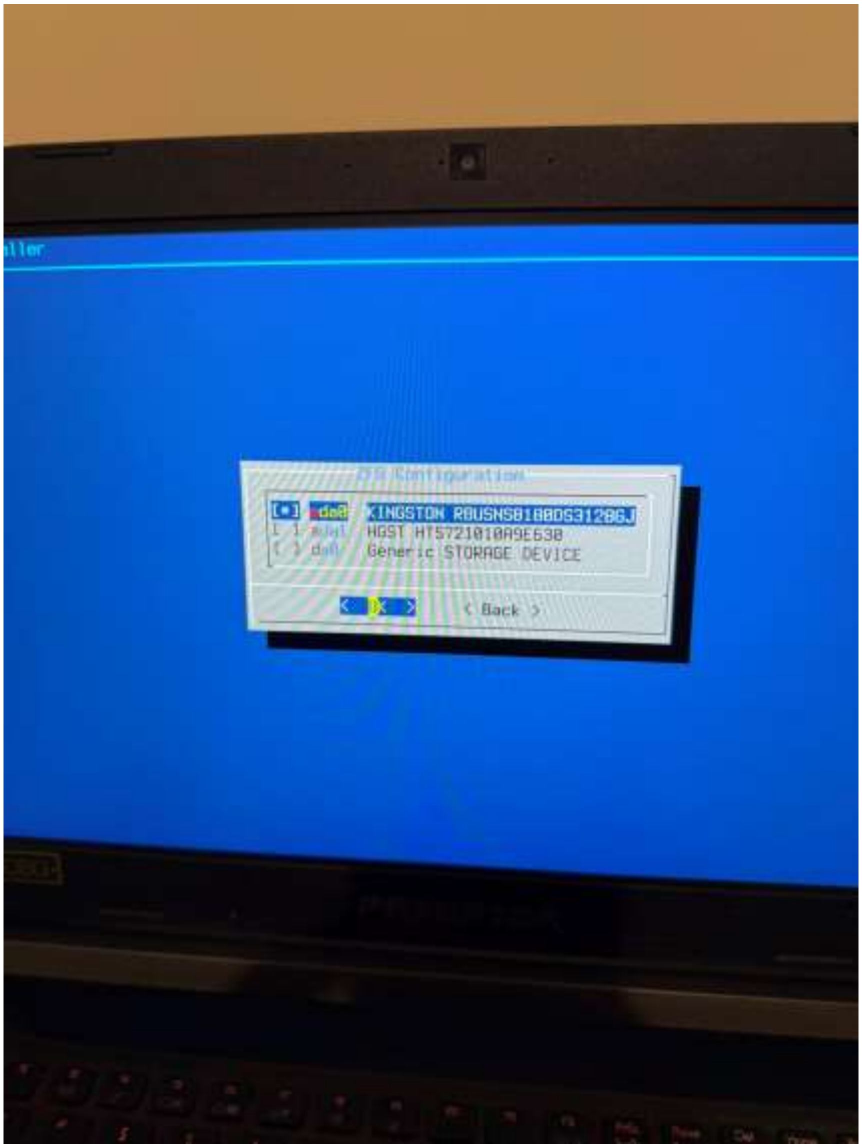
The pfSense installer then showed the copyright and distribution notice. I accepted it and continued to the disk installation steps.



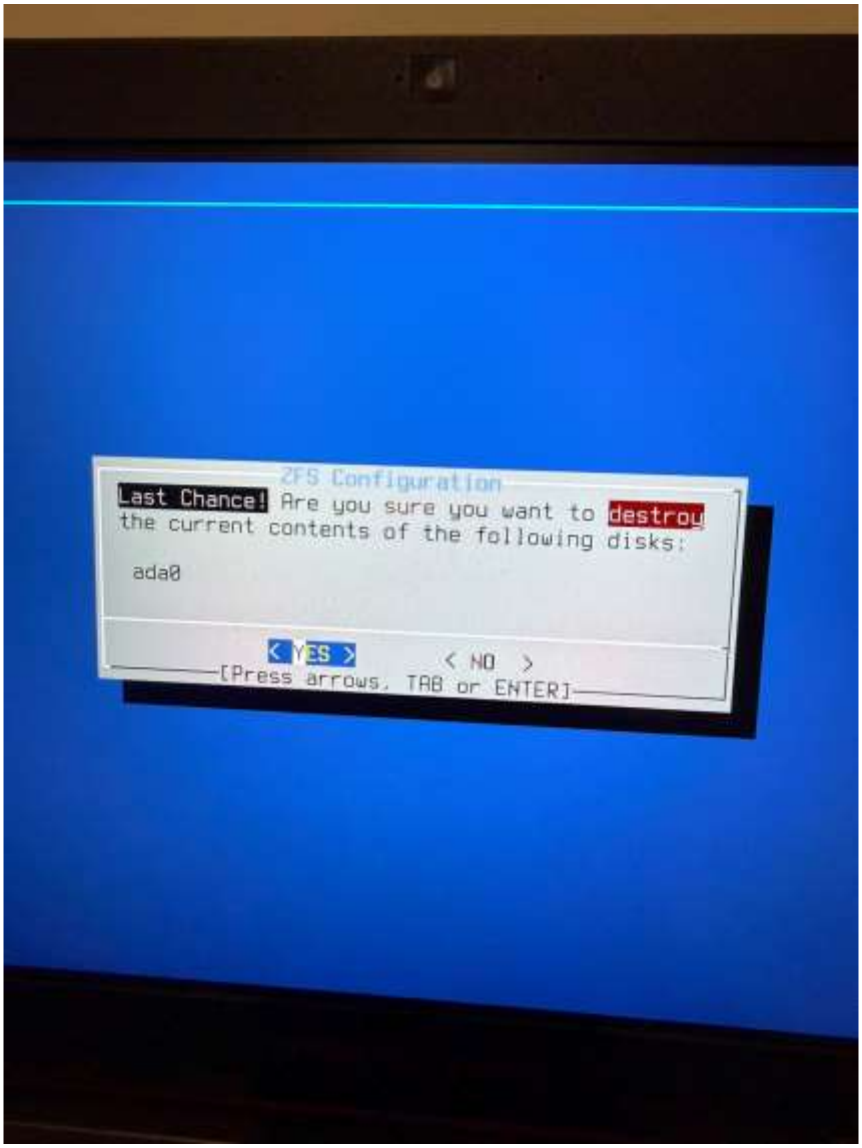
For the installation, I used the ZFS option with a single-disk stripe. I kept the pool name as pfSense and selected the GPT BIOS+UEFI partition scheme.



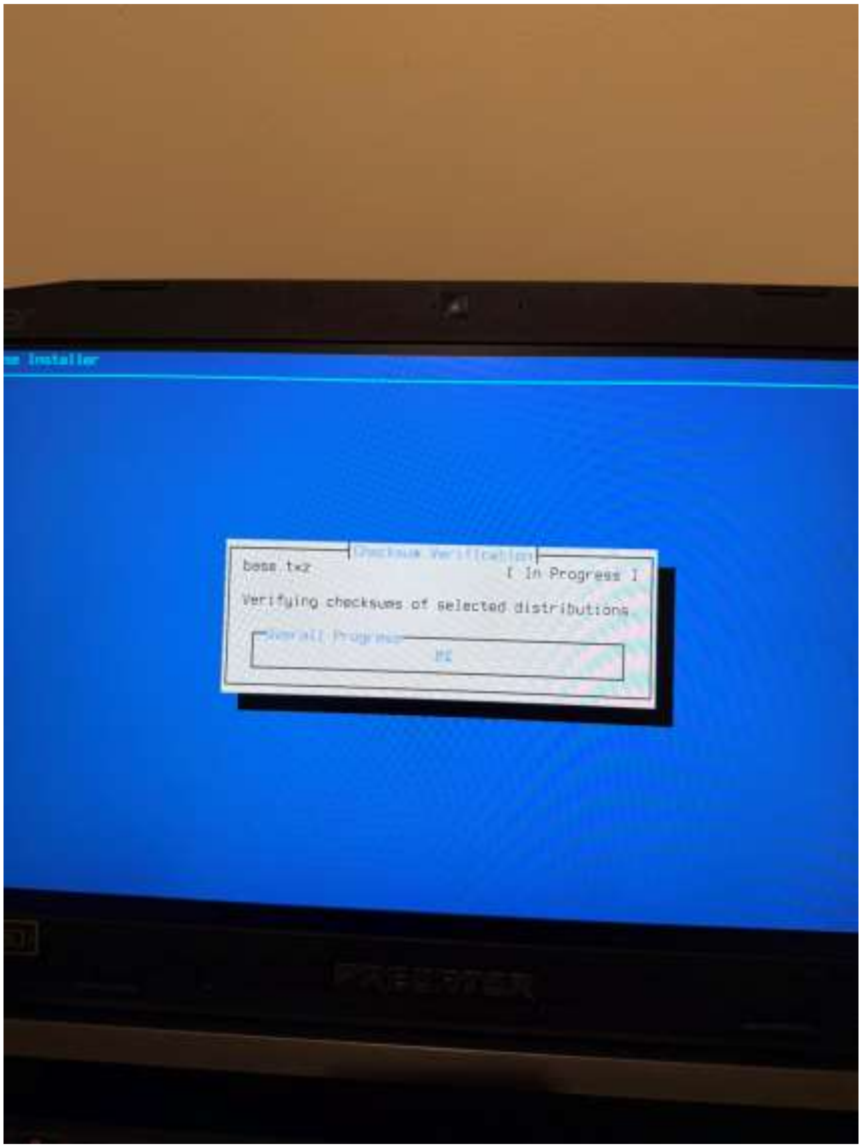
I selected the Kingston SSD as the pfSense installation disk. I did not select the other internal disk or the USB storage device.



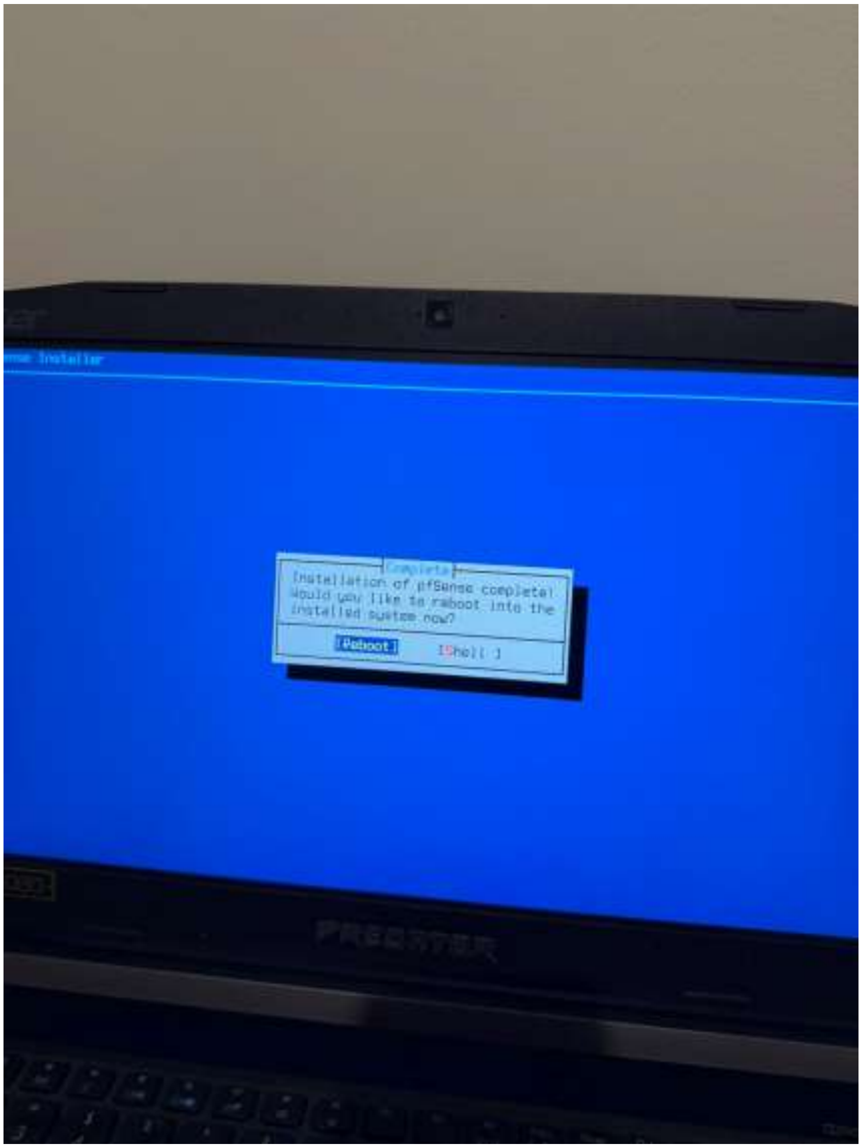
The installer gave a final warning that the selected disk contents would be destroyed. I confirmed the Kingston ada0 disk because it was the disk I selected for the firewall installation.



During installation, pfSense verified the checksums of the selected distribution files before completing the setup.



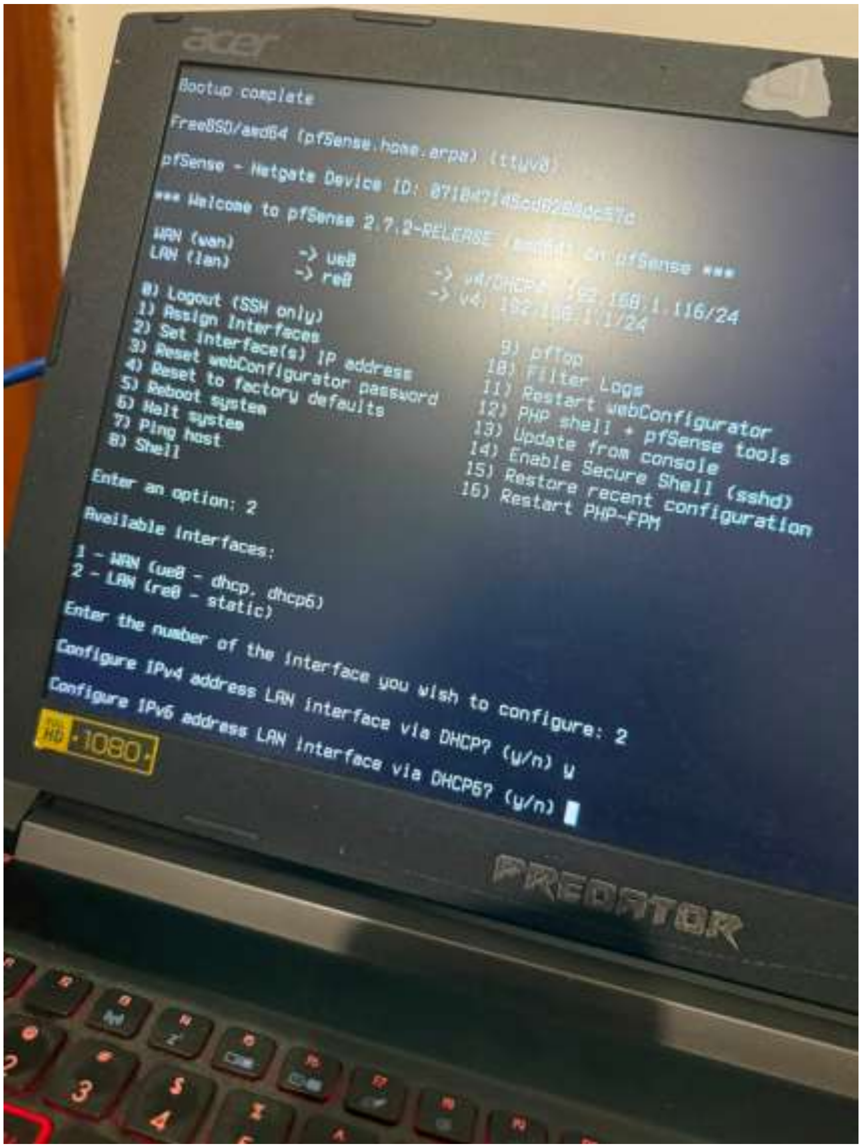
After the installation completed, I selected Reboot to start pfSense from the installed SSD.



After rebooting, pfSense 2.7.2 loaded successfully I assigned the physical interfaces. I selected the USB Ethernet interface ue0 for WAN and the Realtek interface re0 for LAN.

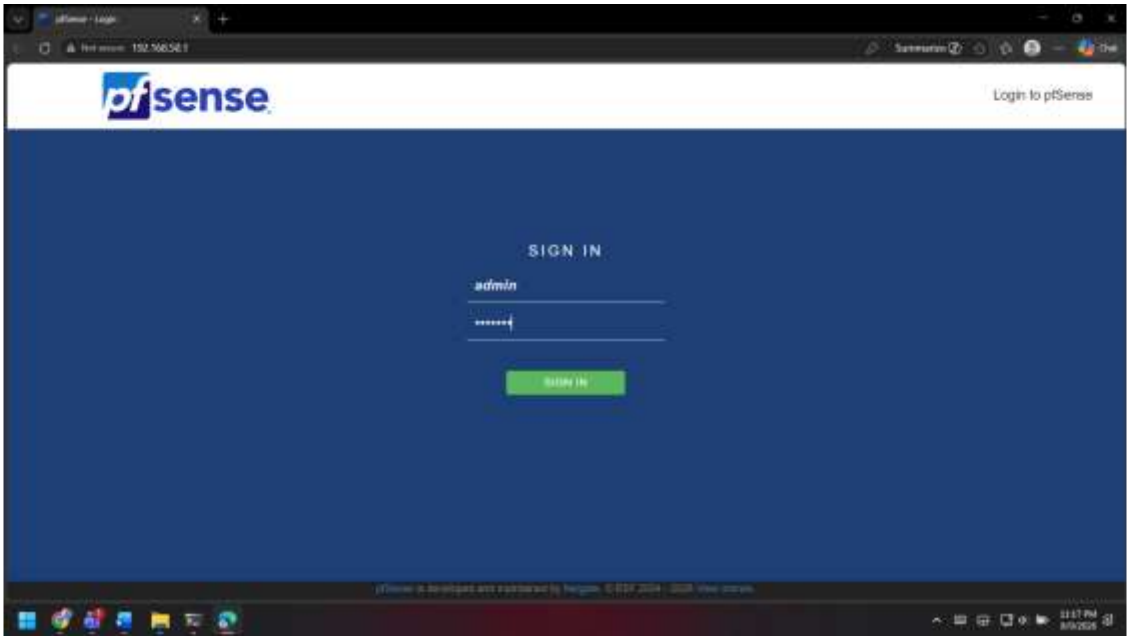
And then it showed the console menu. At this stage, WAN used ue0 and LAN used re0.

Next, I used the console IP configuration option for the LAN interface. This allowed me to move the LAN away from the initial address and prepare a separate network for firewall management.

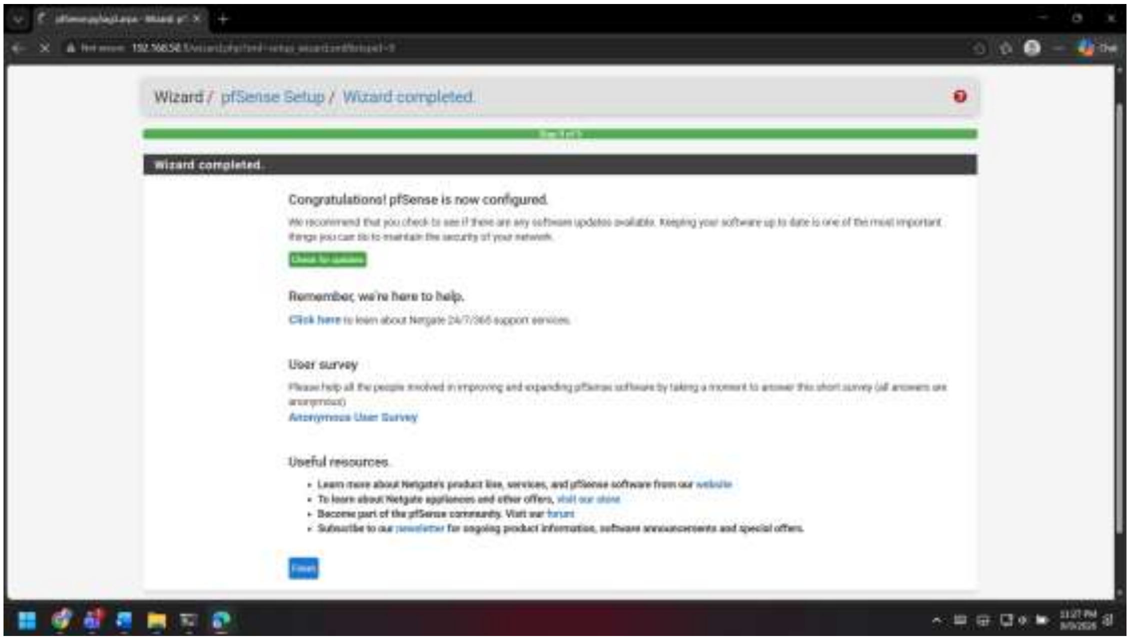


I set the LAN address to 192.168.50.1/24 as having 192.168.1.1/24 was creating problems while going to the pfSense login page as the WAN and LAN were on the same subnet . I also enabled the DHCP server with the client range 192.168.50.100 to 192.168.50.150 so a connected computer could receive an address automatically.

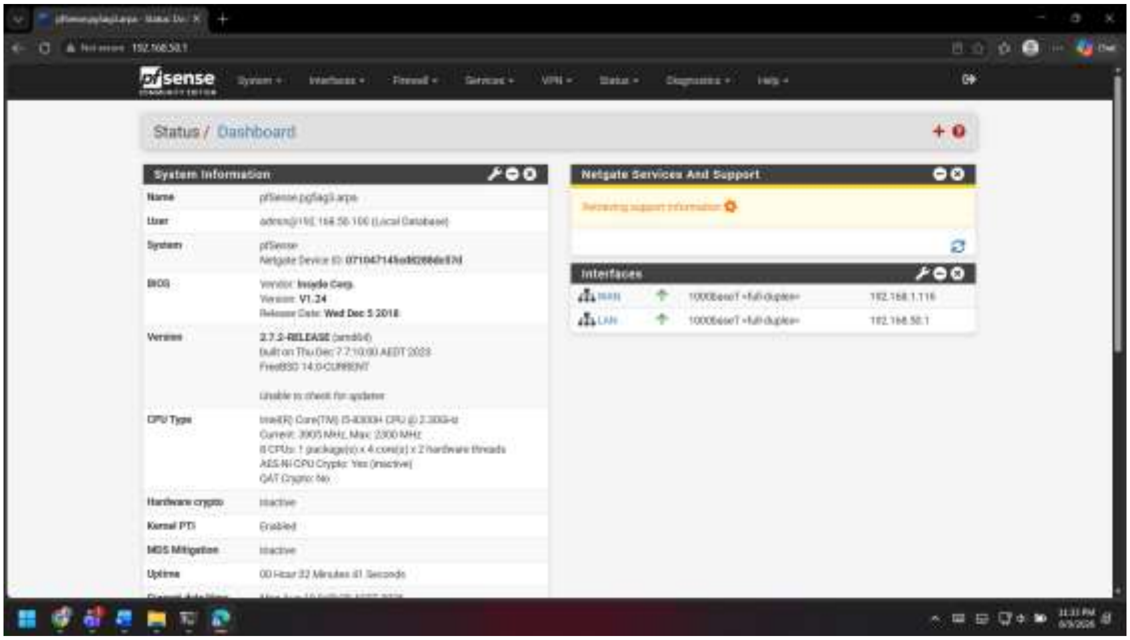
From another computer on the LAN, I opened 192.168.50.1 and reached the pfSense login page. This confirmed that the firewall web interface was reachable from the management computer.



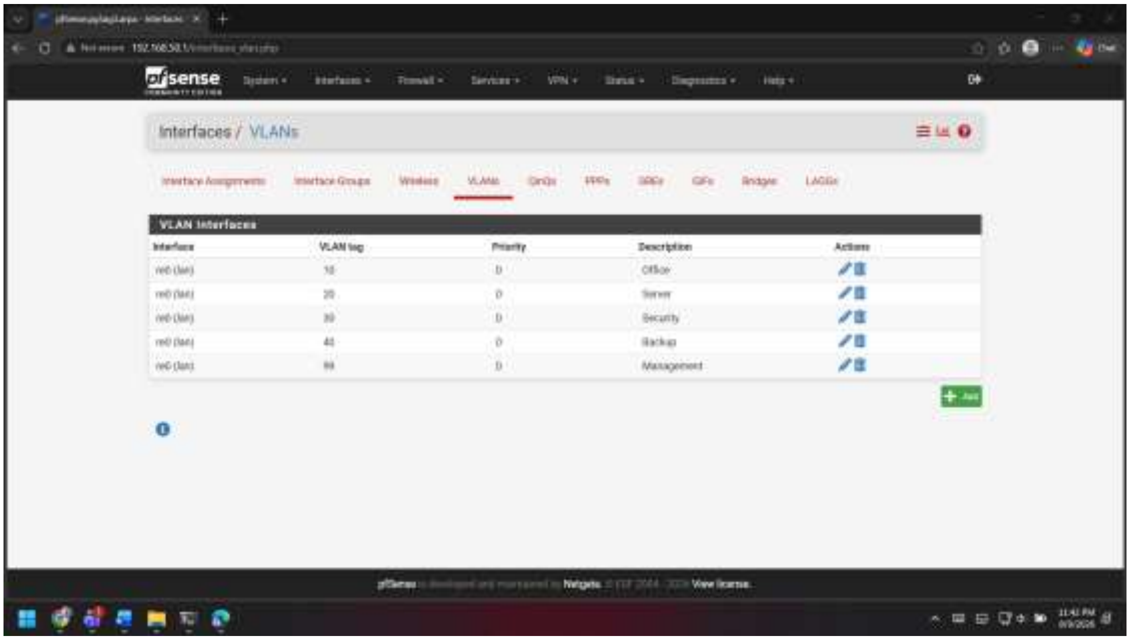
I logged in as the administrator and completed the pfSense setup wizard. The wizard confirmed that the basic pfSense configuration was completed.



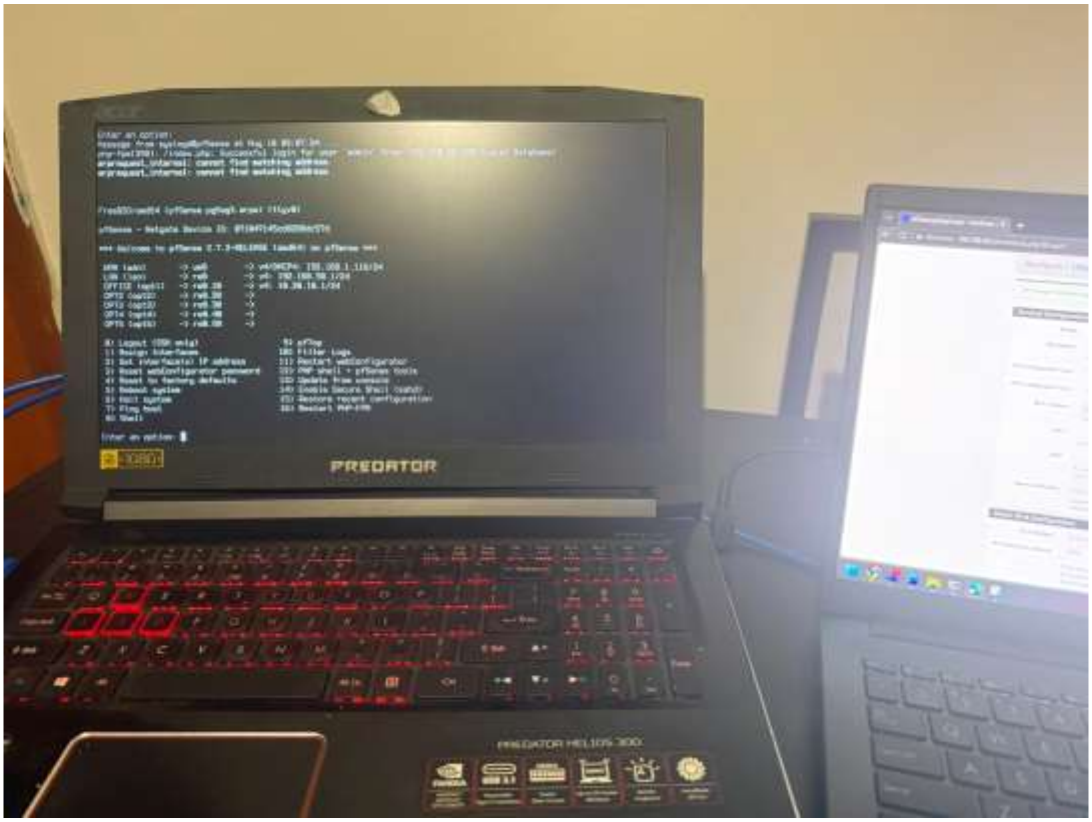
The dashboard then showed pfSense 2.7.2 running with WAN at 192.168.1.116 and LAN at 192.168.50.1. This gave me a working base firewall before adding the project networks.



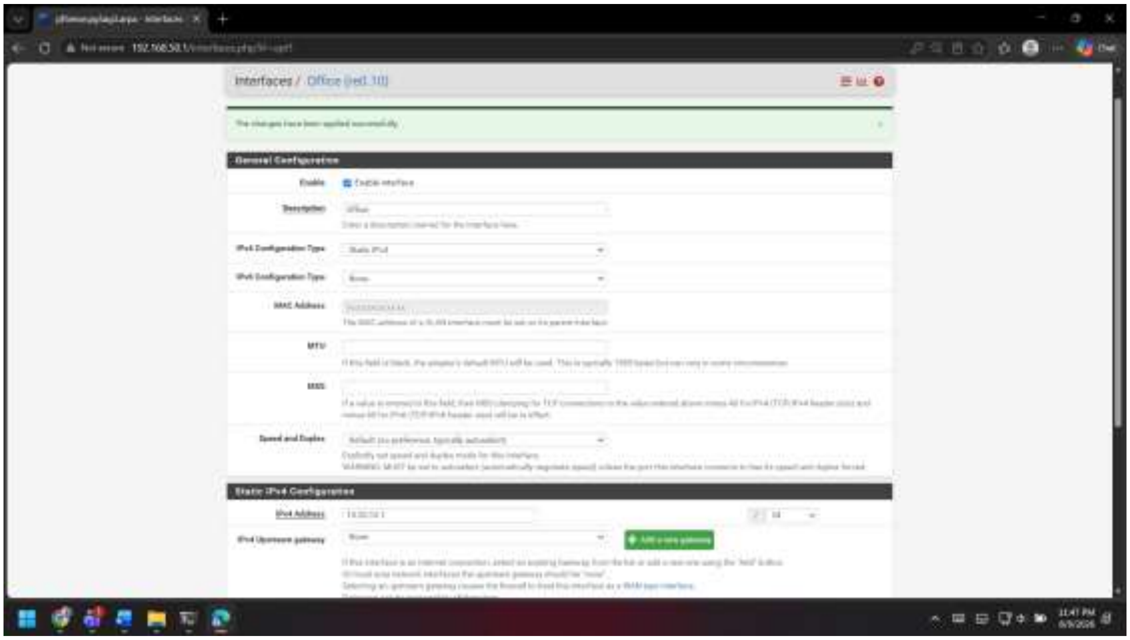
Next, I created the VLANs required for the AG-3 network. I used VLAN 10 for Office, VLAN 20 for Server, VLAN 30 for Security, VLAN 40 for Backup and VLAN 99 for Management.



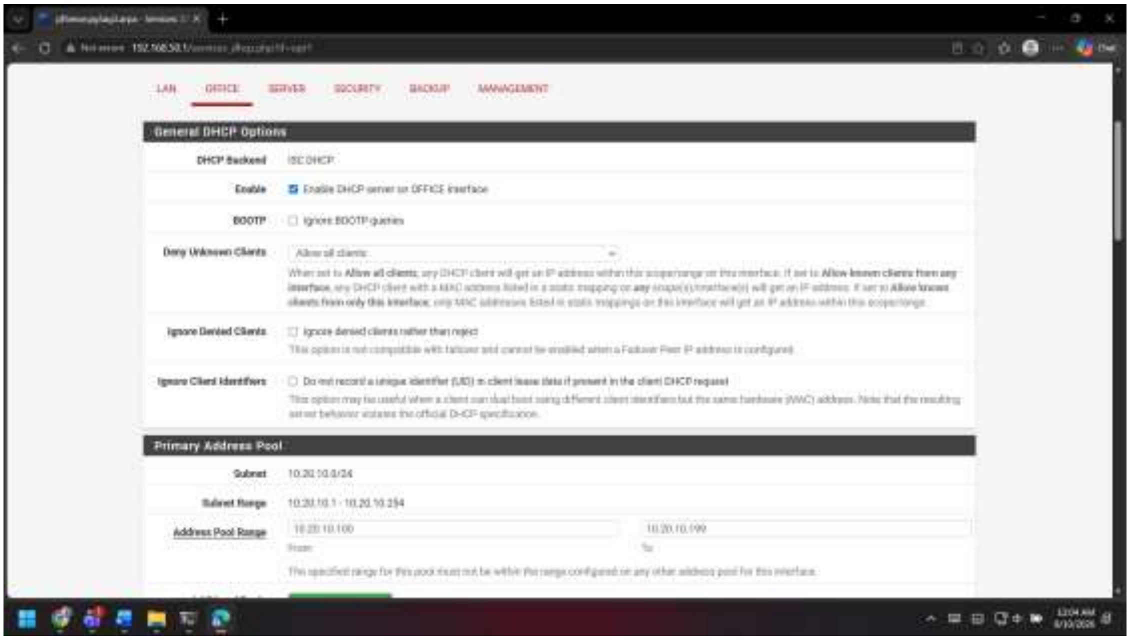
After creating the VLANs, the pfSense console also showed the new VLAN interfaces on re0. This confirmed that I had created the Office, Server, Security, Backup and Management VLAN interfaces on the firewall.



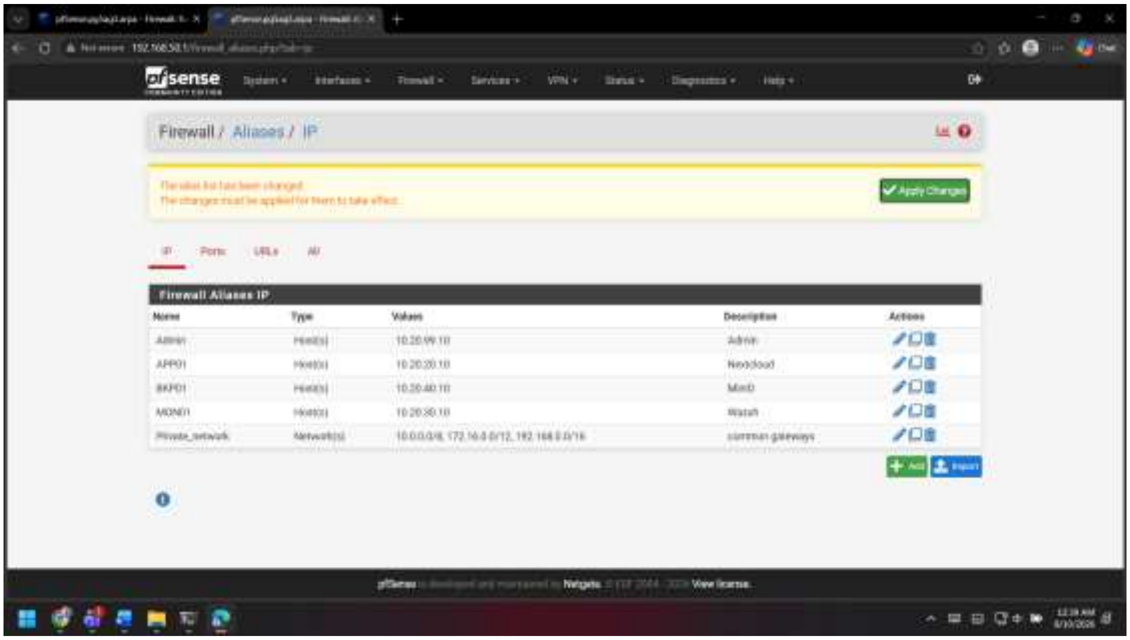
For the Office VLAN, I enabled the interface and configured the static gateway address as 10.20.10.1/24. I then used the same addressing structure as the base for the separated project networks.



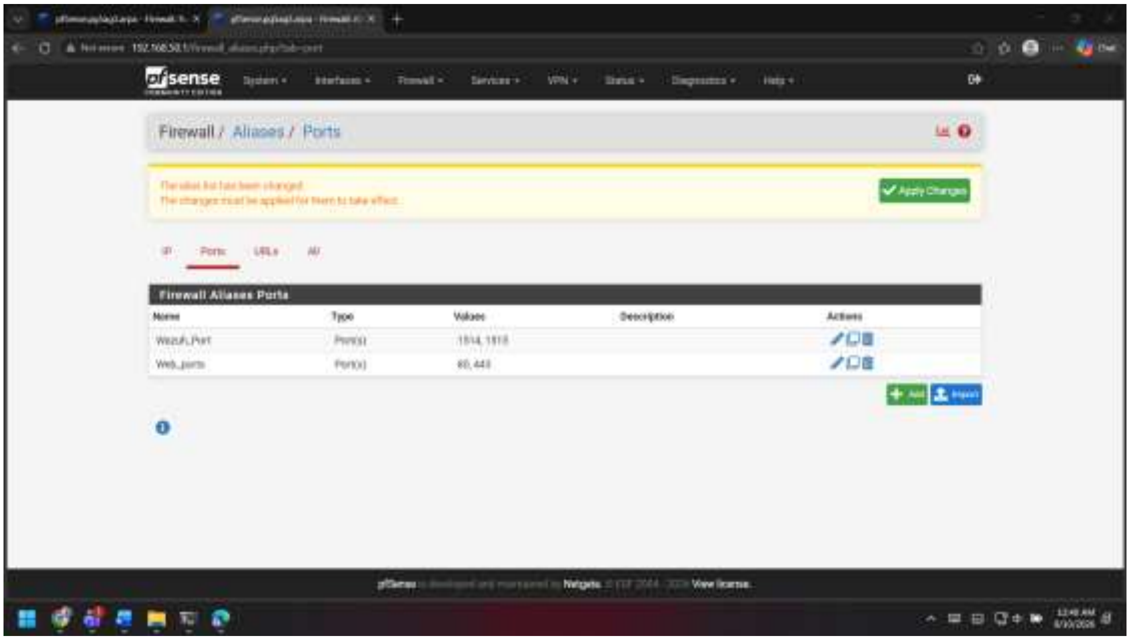
I enabled DHCP on the Office interface and used the address pool 10.20.10.100 to 10.20.10.199. This allows Office devices to receive an IP address from the correct VLAN automatically.



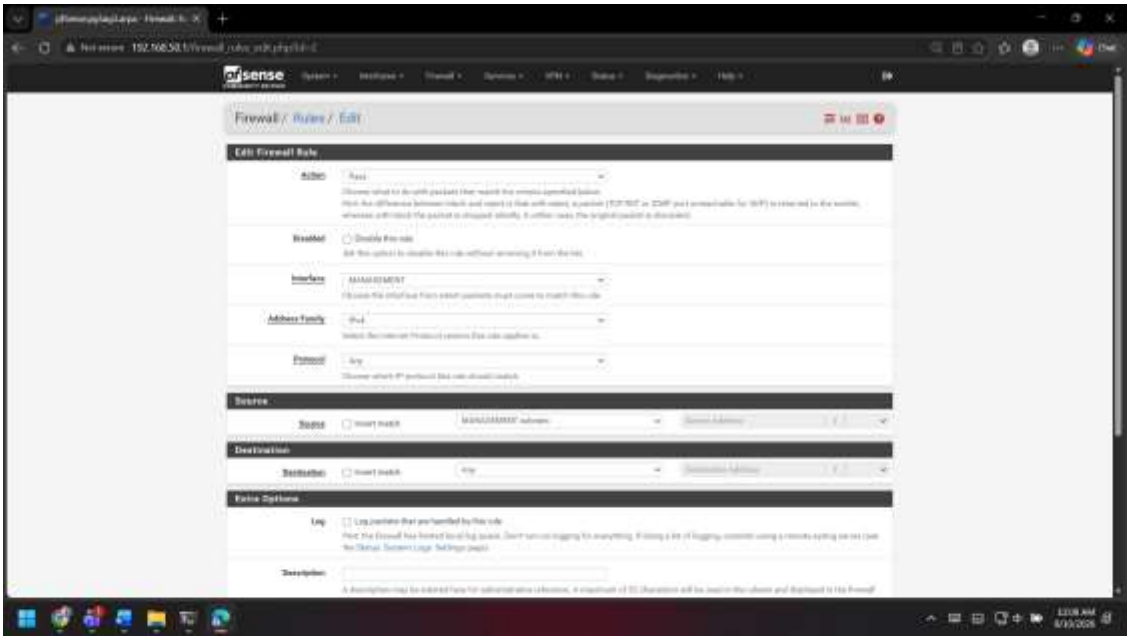
To make the firewall rules easier to manage, I created IP aliases for the main systems. Admin uses 10.20.99.10, APP01 uses 10.20.20.10 for Nextcloud, BKP01 uses 10.20.40.10 for MinIO and MON01 uses 10.20.30.10 for Wazuh. I also created a Private_network alias for the private address ranges.



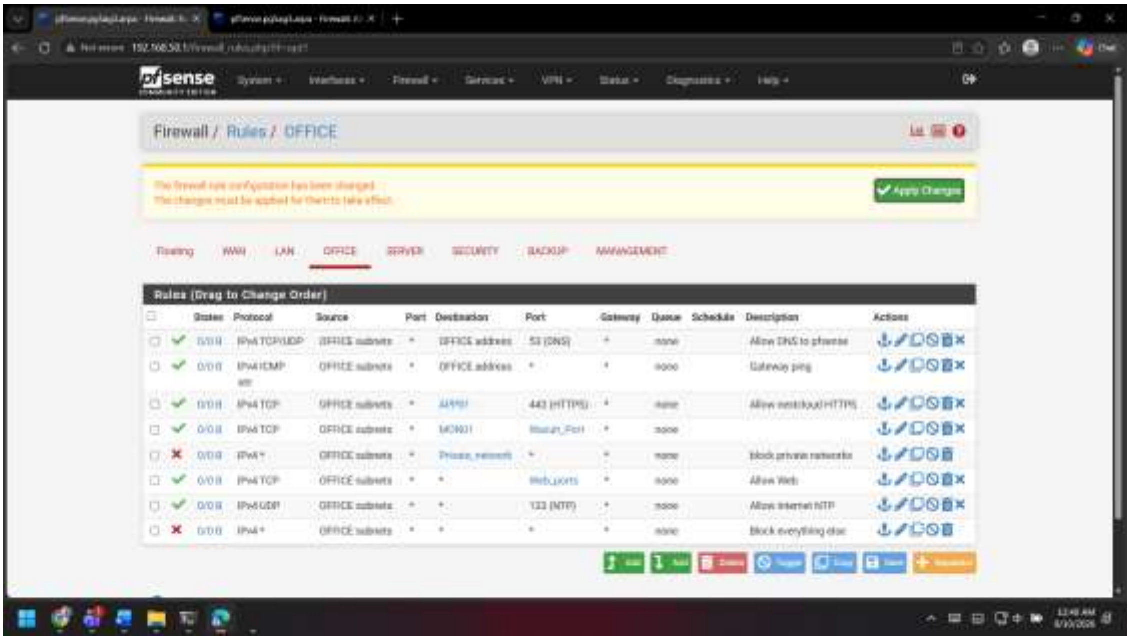
I also created port aliases. Wazuh_Port contains ports 1514 and 1515, while Web_ports contains ports 80 and 443. This makes the firewall rules shorter and easier to understand.



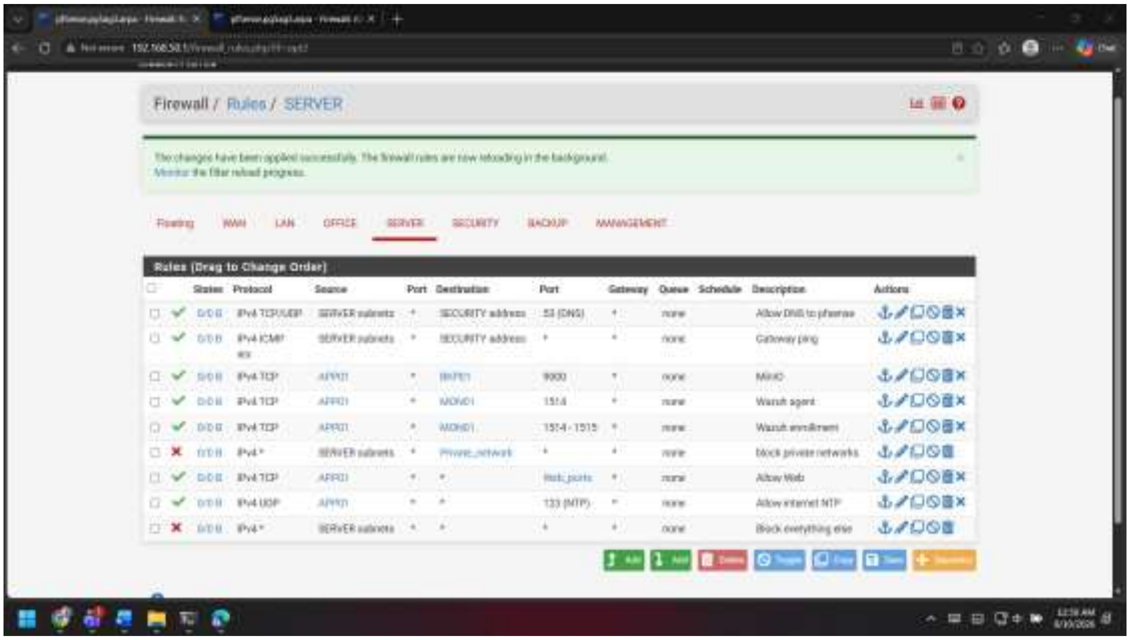
I used the pfSense rule editor to define the interface, protocol, source and destination for each rule. I built the rules separately for every VLAN instead of allowing unrestricted communication between all networks.



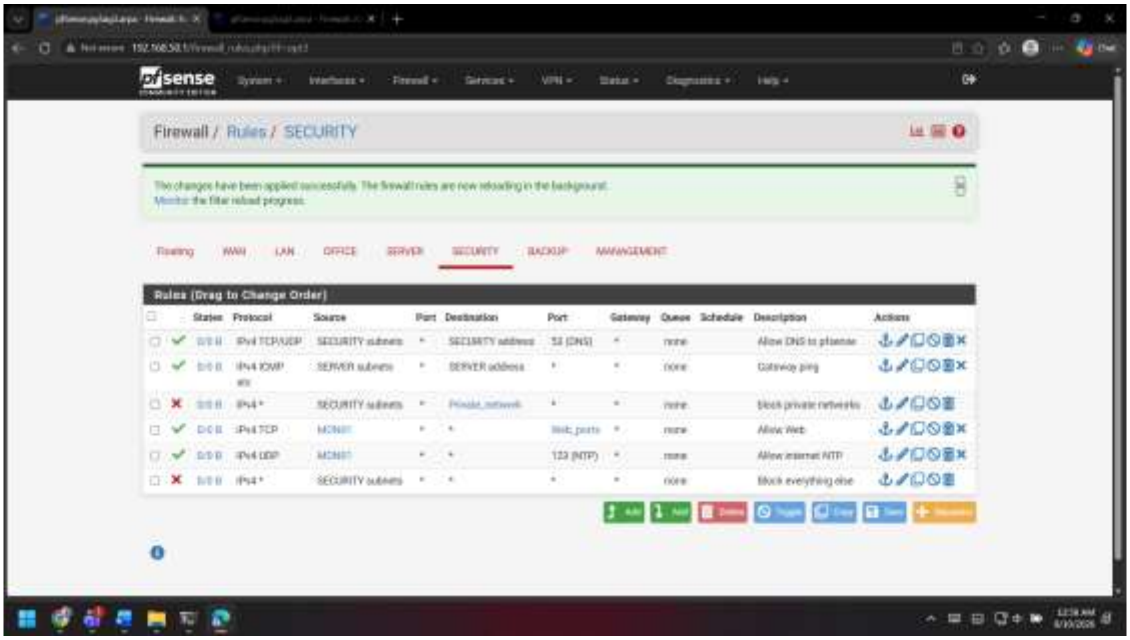
For the Office network, I allowed the required DNS and gateway traffic, Nextcloud HTTPS access, Wazuh communication, normal web access and NTP. I blocked private-network traffic that was not required and finished the rule set with a block-everything-else rule.



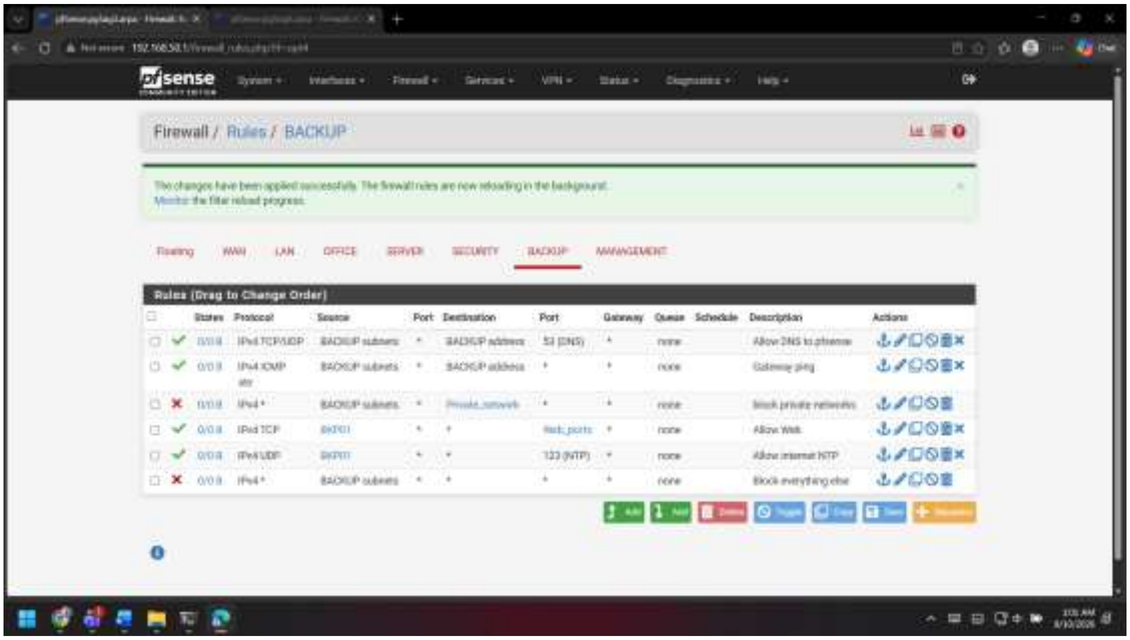
For the Server network, I allowed APP01 to communicate with BKP01 on MinIO port 9000 and with MON01 on the Wazuh ports. I also allowed web and NTP access, while restricting other private-network traffic and finishing with a block-everything-else rule.



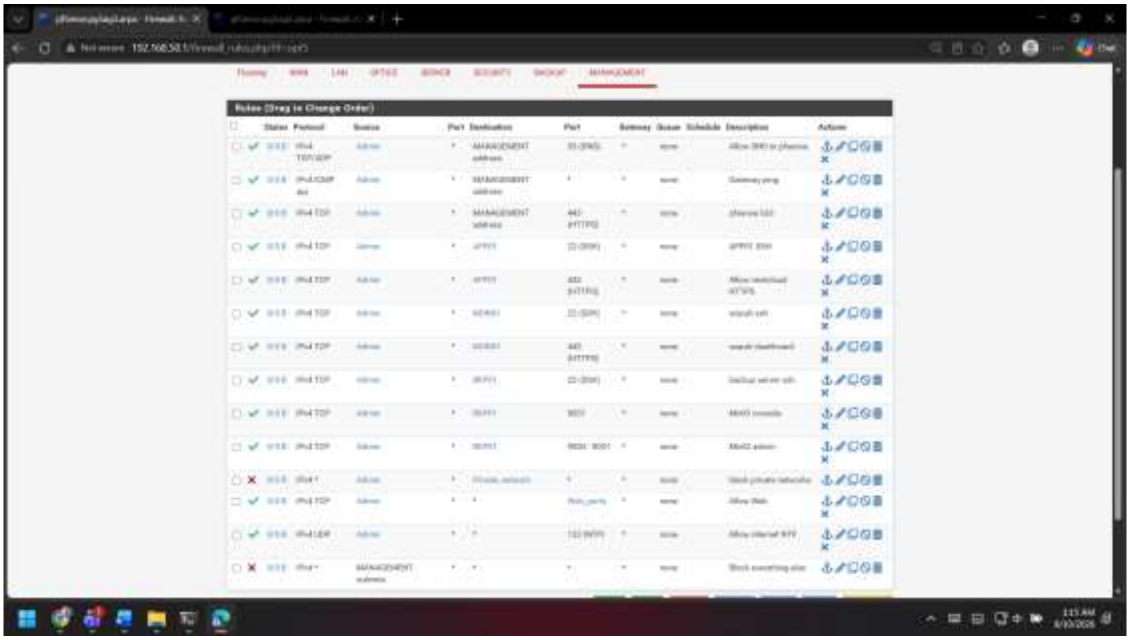
For the Security network, I kept only the required DNS, gateway, web and NTP access for the monitoring system. I also blocked unnecessary private-network traffic and finished with a deny rule for everything else.



For the Backup network, I allowed BKP01 the required web and NTP access. I blocked other private-network access and used the last rule to block any traffic that was not specifically permitted.



Finally, I configured the Management network for the Admin pc. I allowed pfSense GUI access, SSH and HTTPS access to the required servers, Wazuh dashboard access and MinIO administration ports. I also allowed the required web and NTP traffic and finished with a block-everything-else rule.



From this work, I completed the physical pfSense installation, initial interface setup, VLAN segmentation, DHCP configuration, aliases and the first firewall rule set for the AG-3 environment.